

Quoting is Colour-Swap: a model of the rho calculus in the knotted universe

Lucius Gregory Meredith

CEO, F1R3FLY.io, and Chief Science Officer, F1R3FLY Industries

lgreg.meredith@gmail.com

Abstract

The knotted universe of *A Knotted Universe: a new notion of reflective set theories* closes with a conjecture: that the reflective higher-order (rho) calculus has a natural set-theoretic model there, with the quote $@$ read as sealing a process into an atom of the dual colour and the dereference $*$ as unsealing it. We make the conjecture good, and in a form that puts the colour barrier itself to work. We *stratify* the universe into four sorts — red sets and red atoms, black sets and black atoms — linked by the two knots and by a single polymorphic involution, the colour-swap s . The calculus’s reflection operators are then *derived*: $@ = \kappa \circ s$ and $* = s \circ \kappa^{-1}$, so that $*@P \equiv P$ becomes $s \circ s = \text{id}$ — the reflection law is the involutivity of the swap, not an axiom about the model. We give the dynamics as an RSpace semantics: parallel composition is interpreted as keyed multiset union, so the denotation is a monoid homomorphism and the parallel laws of structural congruence become literal equalities of tables — hashtables from channels to polarity-homogeneous bags of produces and consumes, which is exactly the RSpace store. Communication is eager reaction, the zip of dual bags at a key, and it is here, in the zip, that the swap fires, sealing each sent process into the received name; RSpace is the store monad together with this zip, a bialgebra — the precise content of “strictly more than a monad”. Following the GSLT discipline we label transitions not by primitive actions but by *contexts* — the minimal store-insertions enabling a reaction, after Leifer and Milner — and express bisimulation over these context labels; the labels are the program/environment cuts, simultaneously the indices of the OSLF-generated modal logic. The model factors: a resource layer that counts parallel copies, projecting (forget multiplicity and keys) onto the context-behaviour layer. We prove a *full abstraction* theorem at each — $\langle P \rangle \sim_r \langle Q \rangle$ iff $P \sim_r Q$ above, $\langle P \rangle_B = \langle Q \rangle_B$ iff $P \sim Q$ below — by identifying bisimilarity with the kernel of the unique morphism to the final coalgebra; because the labels are contexts, the behavioural equivalence is a congruence by construction, so full abstraction holds for a congruence with no closure step. The risk ledger of the source paper transfers: finite processes cost no infinite risk; recursion — which in rho enters through reflection, not a primitive replication — costs one ω , black infinity, and not one primitive name. Aczel’s CCS model is recovered as the equivariant, colour-respecting fragment.

1 Introduction

Aczel modelled Milner’s CCS inside non-well-founded set theory [1]: a process P is identified with the set of its labelled transitions $\{(a, P') : P \xrightarrow{a} P'\}$, a looping process is a non-well-founded set, bisimilar processes are *equal*, and the anti-foundation axiom’s solution lemma individuates processes as the unique solutions of their behavioural equations. The construction is canonical, but it buys two things by fiat: an external, unstructured supply of action names a , and the non-well-foundedness itself.

The reflective higher-order calculus — the rho calculus [3] — differs from CCS and the π -calculus precisely where Aczel spends his first postulate. Rho has *no* primitive names. A name is a quoted process: the quote $@$ turns a process into a name, the dereference $*$ turns a name back into the process it quotes, and the two compose to the identity, $*@P \equiv P$. There is correspondingly no restriction operator ν and no scope extrusion: fresh names are *computed*, by quoting.

The knotted universe of [2] was built from the empty set alone, as two mutually recursive copies of set theory — a red copy $\text{Set}[X]$ and a black copy $\text{Set}[\bar{X}]$ — tied by $X = \text{Set}[\bar{X}]$, $\bar{X} = \text{Set}[X]$, so that each colour’s *atoms* are the other colour’s *sets*. Section 9.3 of [2] conjectures this universe is the natural home for an Aczel-style model of rho, with $@$ sealing a process into an atom of the opposite colour and $*$ unsealing it. We carry the conjecture out. Three design decisions shape the development.

Quoting is the colour-swap, read through a stratification (Section 3). Rather than collapse “red atom” and “black set” into one sort, as the bare knot does, we keep four sorts and make the colour-moving maps first class. A single polymorphic involution $\mathbf{s}$, the colour-swap, exchanges red and black *while preserving the set/atom distinction*; the two knots $\kappa, \bar{\kappa}$ supply the set/atom coercions. The calculus’s $@$ and $*$ are then *composites* of these — $@ = \kappa \circ \mathbf{s}$, $* = \mathbf{s} \circ \kappa^{-1}$ — and the reflection law is $\mathbf{s} \circ \mathbf{s} = \text{id}$. This is at once more honest (the colour crossing is explicit, not hidden in a definitional identity) and exactly the graded, colour-tracking discipline [2], §6.5 asks for: the four sorts are the grading, and $\mathbf{s}$ is the reflective move.

The dynamics are an RSpace semantics (Section 5). Modulo structural congruence the parallel fragment is the free commutative monoid, hence the finite-multiset construction; interpreting $|$ as multiset union makes the denotation a monoid homomorphism on the nose, so the parallel laws of $\equiv$ become literal equalities. Bucketing the multiset by channel yields a hashtable from names to polarity-homogeneous bags of produces and consumes — precisely the RSpace store. Following Milner’s abstraction/concretion reading [4], inputs install consume bags of abstractions $(y)P$ and outputs produce bags of data Q ; communication is *eager* reaction — when a key carries both polarities, zip the bags and fire each matched pair by application $(y)P \cdot Q = P\{Q/y\}$, the surplus remaining — and it is in this zip that the swap of Section 3 fires, sealing the sent process into the received name. RSpace is the store monad together with the zip: a bialgebra, which is the precise sense in which it is “more than a monad”. The model factors — the table projects, forgetting multiplicity and keys, onto the context-behaviour universe of the previous sections — so the resource-sensitive RSpace layer sits above the bisimulation layer and maps onto it.

Transitions are labelled by contexts (Sections 2.3, 6). To stay consistent with the GSLT programme we do not label transitions by primitive actions but by *contexts*: $P \xrightarrow{F} P'$ means F is a minimal environment — an idem-pushout, after Leifer and Milner [5, 6], adapted to GSLTs [7] — in which P completes a reaction to P' . The label is the program/environment cut at which P meets its world, and it is at once the Firing label of the transition system, the Witnessing index of an OSLF-generated Hennessy–Milner modality $\langle F \rangle \varphi$, and the Placing of the interaction in P ’s term structure — three views of one ∂T . Expressing bisimulation over context labels has a structural dividend: the equivalence is a congruence by construction, so the full-abstraction theorem holds for a congruence with none of the closure-under-substitution care an action-labelled account requires.

Recursion is free (Sections 4, 6). The denotation lands in the final coalgebra of a behaviour functor over the knotted nominal universe; the colour-blind non-well-foundedness of [2], §7 supplies

the recursion (which in rho is reached through reflection, with no primitive replication operator), and finality makes bisimilar processes equal. We prove full abstraction, $\llbracket P \rrbracket = \llbracket Q \rrbracket \iff P \sim Q$, identifying context bisimilarity with the kernel of the unique morphism to the final coalgebra, and read it back through OSLF adequacy.

We work throughout with the modern presentation: $@P$ for quoting, $*x$ for dereference, $\text{for}(y \leftarrow x) P$ for input.

2 The rho calculus

2.1 Syntax

The two-sorted grammar of processes P and names x :

$$P, Q ::= \mathbf{0} \mid P \mid Q \mid \text{for}(y \leftarrow x) P \mid x!(Q) \mid *x, \quad x, y ::= @P.$$

$\mathbf{0}$ is the stopped process; $P \mid Q$ parallel composition; $\text{for}(y \leftarrow x) P$ the input receiving on channel x , binding the received name to y in P ; $x!(Q)$ the asynchronous output sending Q on x ; and $*x$ the dereference. A name is solely a quoted process: no primitive names, and no restriction ν . In $\text{for}(y \leftarrow x) P$ the name y is bound in P . Capture-avoiding substitution $P\{x/y\}$ of a name for a free name reaches inside quotes.

There is no primitive replication or recursion. Both are nonetheless available, by reflection: a recursive process is written by quoting its own continuation and dereferencing it, e.g. a server $D \equiv \text{for}(y \leftarrow x) (*y \mid x!(y)) \mid x!(@(\dots))$ reinstalls itself after each interaction. Infinite behaviour from a finite term is therefore present already in the core calculus — which is why the model needs the non-well-founded reading even with no $!$ in the grammar.

2.2 Structural congruence and name equivalence

$\equiv$ is the least congruence closed under α -equivalence and

$$(P \mid Q) \mid R \equiv P \mid (Q \mid R), \quad P \mid Q \equiv Q \mid P, \quad P \mid \mathbf{0} \equiv P, \quad *@P \equiv P.$$

Name equivalence is induced by dereference: $@P \equiv_{\mathcal{N}} @Q :\iff *@P \equiv *@Q \iff P \equiv Q$.

2.3 Reaction and the context-labelled transition system

The base interaction is communication, with the structural propagation rules:

$$\frac{x \equiv_{\mathcal{N}} z}{\text{for}(y \leftarrow x) P \mid z!(Q) \longrightarrow P\{@Q/y\}} \text{COMM} \qquad \frac{P \longrightarrow P'}{P \mid R \longrightarrow P' \mid R} \text{PAR}$$

$$\frac{P \equiv P' \quad P' \longrightarrow Q' \quad Q' \equiv Q}{P \longrightarrow Q} \text{STRUCT}$$

Rho is an interactive GSLT whose single base constructor is parallel composition: $K = \mid$ is the head of the left-hand side of the one hypothesis-free rule, COMM, and the cut it induces is *relative* — each parallel component is, from the other's vantage, the environment.

To define bisimulation we follow the GSLT discipline and label transitions by *contexts* rather than by primitive actions, after the minimal-context (relative-pushout) construction of Leifer and Milner [5, 6], as adapted to GSLTs [7]. A *context* F is a process with one hole $[\cdot]$; $F[P]$ is the

result of plugging P . Communication never fires under a prefix, so the reactive contexts are, modulo $\equiv$, the parallel contexts $[\cdot] \mid R$: the cut always lies at a parallel composition, exactly the program/environment cut.

Definition 1 (Context-labelled transitions). $P \xrightarrow{F} P'$ holds when F is a context with $F[P] \longrightarrow P'$ by a single COMM, and F is *minimal* — an idem-pushout, supplying just the missing half of the redex and no more. Concretely, modulo $\equiv$:

- (*offer input*) if $P \equiv \text{for}(y \leftarrow x) P_0 \mid P_1$ then $P \xrightarrow{[\cdot] \mid x!(Q)} P_0\{\text{@}Q/y\} \mid P_1$, one transition per message Q ;
- (*offer output*) if $P \equiv x!(Q) \mid P_1$ then $P \xrightarrow{[\cdot] \mid \text{for}(y \leftarrow x) R} P_1 \mid R\{\text{@}Q/y\}$, one per input context $\text{for}(y \leftarrow x) R$;
- (*internal redex*) if $P \equiv \text{for}(y \leftarrow x) P_0 \mid x!(Q) \mid P_1$ then $P \xrightarrow{[\cdot]} P_0\{\text{@}Q/y\} \mid P_1$.

The label records what environment P needs to interact; the identity context $[\cdot]$ labels the reactions P performs alone (the silent moves). The seal $\text{@} = \kappa \circ \mathbf{s}$ acts in each substitution: communication delivers the *name* $\text{@}Q$ of the sent process, never the process itself.

Remark 2 (The label set is ∂T , in its three views). The labels are the Fire-sub-bundle ∂T_K of the splitting space $\partial T \times T$: the splittings (F, t) whose subterm t completes a COMM redex. This one set is read three ways [7]: as the label set of the transition system (*Firing*); as the index set of the OSLF-generated Hennessy–Milner modalities $\langle F \rangle \varphi$ — “some F -labelled transition reaches a state satisfying φ ” (*Witnessing*); and as the program/environment cuts, i.e. the locations at which P meets its environment (*Placing*). We will use the first to define bisimulation and the second to characterise it.

Definition 3 (Context bisimulation). A symmetric relation $\mathcal{R}$ on processes is a *context bisimulation* if $P \mathcal{R} Q$ implies: whenever $P \xrightarrow{F} P'$ there is $Q \xrightarrow{F} Q'$ with $P' \mathcal{R} Q'$ (and symmetrically). Context bisimilarity $\sim$ is the largest context bisimulation; it contains $\equiv$.

Proposition 4 (Congruence, by construction). *If the GSLT minimal-context construction supplies relative pushouts for rho’s contexts [5, 6, 7], then context bisimilarity is a congruence: $P \sim Q$ implies $F[P] \sim F[Q]$ for every context F .*

Proof sketch. This is the Leifer–Milner theorem: when reactions are presented as idem-pushout labels over a category of contexts possessing relative pushouts, the derived bisimilarity is a congruence, because the bisimulation game already quantifies over the discriminating power of every context. The hypothesis — that rho’s contexts have the requisite relative pushouts — is the content of the GSLT minimal-context construction [7], and is recorded as Obligation 5. $\square$

This is the structural reason the GSLT programme labels by contexts: congruence is *built into* the equivalence rather than recovered afterwards. It settles, at the level of the definition, the point that an action-labelled presentation must negotiate by closing under name instantiation (compare Section 6).

Remark 5 (Commitments are the finite presentation). The offer-input clause of Definition 1 is a family $\{P \xrightarrow{[\cdot] \mid x!(Q)} P_0\{\text{@}Q/y\} \mid P_1 : Q \in \mathbf{Proc}\}$ indexed by all messages. The family factors through a single *abstraction* $(y)(P_0 \mid P_1)$ (with $y \# P_1$): plugging the message $\text{@}Q$ into the abstraction gives the residual. Likewise the offer-output family factors through a *concretion*. These abstraction/concretion packagings, in Milner’s sense [4], are the finite presentation of the class-indexed context-transition families; we use them for the denotation of Section 5, with the understanding that the labels they abbreviate are the contexts of Definition 1.

3 Quoting is the colour-swap: a stratified reading

3.1 Four sorts and one involution

The construction of [2] takes two copies of one signature-generated set theory, red $\text{Set}[X]$ and black $\text{Set}[\bar{X}]$, and ties them by $X = \text{Set}[\bar{X}]$, $\bar{X} = \text{Set}[X]$: each colour’s atoms are the other’s sets. The source paper reads these ties as identities, collapsing “red atom” with “black set”. For the rho model it is cleaner — and, as we will see, exactly the graded discipline [2], §6.5 calls for — to keep the sorts distinct and make the colour-moving maps first class. We name four sorts,

$$\mathbf{S} \text{ (red sets), } \mathbf{A} \text{ (red atoms), } \mathbf{S} \text{ (black sets), } \mathbf{A} \text{ (black atoms),}$$

and three families of maps.

The knots, as coercions. The two recursive knots become two isomorphisms,

$$\begin{aligned} \kappa: \mathbf{S} &\xrightarrow{\cong} \mathbf{A} && \text{(black sets are red atoms; } X = \text{Set}[\bar{X}]), \\ \bar{\kappa}: \mathbf{S} &\xrightarrow{\cong} \mathbf{A} && \text{(red sets are black atoms; } \bar{X} = \text{Set}[X]). \end{aligned}$$

Where [2] writes $X = \text{Set}[\bar{X}]$ as an equality we write κ as a coercion: a black set, presented *as* a red atom, is red-opaque ([2], Remark 2); κ is the act of so presenting it.

The swap, an involution. The two copies are copies of one theory, so the construction is symmetric under exchanging the colours — the symmetry [2] invokes for its Corollary 2. We make it a map: a single polymorphic, *sort-preserving* involution

$$\mathbf{s}: \mathbf{S} \cong \mathbf{S}, \quad \mathbf{s}: \mathbf{A} \cong \mathbf{A}, \quad \mathbf{s} \circ \mathbf{s} = \text{id},$$

repainting red to black and back, structure for structure ($\mathbf{s}$ carries $\emptyset$ to $\emptyset$, $\cup$ to $\cup$, and so on). The swap commutes with the knots: the square

$$\mathbf{s} \circ \kappa = \bar{\kappa} \circ \mathbf{s} \quad (\text{both } \mathbf{S} \rightarrow \mathbf{A})$$

holds, expressing that repainting a black-set-as-red-atom agrees with repainting then re-presenting. This coherence is what lets one colour’s reflection generate the other’s.

3.2 The calculus’s reflection, derived

Take processes to be red sets and names to be red atoms. Define

$$\boxed{\text{@} := \kappa \circ \mathbf{s} : \mathbf{S} \rightarrow \mathbf{A}, \quad * := \mathbf{s} \circ \kappa^{-1} : \mathbf{A} \rightarrow \mathbf{S}.}$$

Thus $\text{@}P$ first repaints the red process P black ($\mathbf{s}$), then presents the result as a red atom (κ): it “seals P into an atom of the opposite colour” exactly as [2], §9.3 proposes, the seal being the red-opacity of κ , the colour crossing being $\mathbf{s}$. The dereference $*x$ un-presents (κ^{-1}) and repaints back ($\mathbf{s}$).

Proposition 6 (Reflection is involutivity of the swap). *For every process P and name x ,*

$$*\text{@}P = \mathbf{s} \kappa^{-1} \kappa \mathbf{s}(P) = \mathbf{s} \mathbf{s}(P) = P, \quad \text{@}*x = \kappa \mathbf{s} \mathbf{s} \kappa^{-1}(x) = x.$$

Hence the model validates $\text{@}P \equiv P$ and $\text{@}*x \equiv_{\mathcal{N}} x$, not by imposing an equation but as the assertion that the colour-swap is an involution.*

Proposition 7 (Names implemented by swaps; the dual calculus). *Define the black calculus's reflection by $@_S := \bar{\kappa} \circ s : S \rightarrow A$ and $*_S := s \circ \bar{\kappa}^{-1}$. Then by the coherence $s\kappa = \bar{\kappa}s$,*

$$@_S = s \circ @ \circ s,$$

so the two colours' quotes are swap-conjugate: one reflection determines the other. The black calculus has black sets as processes and red sets as names; by [2], Corollary 2 the two interlock, each calculus's names being the other's processes. A single reflective universe carries two dual rho calculi at once.

Remark 8 (The stratification is the grading). [2], §6.5 calls for a graded modality separating colour-respecting (equivariant) computation from reflective computation, the price of breaking a name's opacity. The four sorts are that grading. A computation that stays in $\{S, A\}$ and uses only $@, *$ relabels red processes and names without ever observing black structure: this is the equivariant fragment, ordinary nominal/ π -style names with every guarantee intact. Applying s to compute with black set-operations on a name's interior is the reflective surplus, and the type — which sort one is in — records it. Decidable structural identity of names ([2], §6.5) is here the decidability of equality in A , settled out of red's sight by black equality of the swapped processes.

4 The behaviour functor and the process universe

The knotted universe is a model of nominal sets ([2], §6.1): the model M of its Theorem 1 carries the $\text{Sym}(\mathcal{A})$ -action with every element finitely supported and supports the name-abstraction $[\mathcal{N}]X = (\text{Name} \times X)/\sim$ with $\text{supp}((a)x) = \text{supp}(x) \setminus \{a\}$ ([2], §6.2). Write $\mathcal{U}$ for this nominal universe of red sets and identify $\text{Name} = A$ with $\text{Proc} = S$ along the reflection of Section 3.

Definition 9 (Behaviour functor). Let $B : \mathcal{U} \rightarrow \mathcal{U}$,

$$B(R) = \mathcal{P}(\partial T_K \times R),$$

the (powerset of) context-label/residual pairs of Definition 1, with ∂T_K the Fire context-labels (Remark 2). Because the input and output labels carry an unrestricted message or continuation, ∂T_K is class-indexed and B is taken with the full powerclass $\mathcal{P}$ rather than $\mathcal{P}_{\text{fin}}$; its final coalgebra accordingly lives in the class-sized framework of [2], §8.1, where it coincides with the colour-blind non-well-founded universe of [2], §7. The *commitment refinement* $B^b(R) = \mathcal{P}_{\text{fin}}(\text{Name} \times [\mathcal{N}]R \uplus \text{Name} \times \text{Name} \times R \uplus R)$ (input abstractions, output concretions, silent moves) presents the same bisimilarity finitely (Remark 5), assembled from constants, products, coproducts, finite powerset, and the abstraction functor $[\mathcal{N}](-)$, each preserving finite support and weak pullbacks; we use B^b when finite branching matters.

A B -coalgebra (R, β) equips every r with its set of context-labelled transitions — the data of Section 2.3. A coalgebra morphism is a finitely-supported map commuting with the structure maps: a functional context bisimulation.

Definition 10 (The process universe). $\text{Proc} := \nu B$, the final B -coalgebra in $\mathcal{U}$, with structure map $\langle \text{out}, \text{in}, \tau \rangle : \text{Proc} \rightarrow B(\text{Proc})$, and $\text{Name} := s(\text{Proc})$ presented as A via κ .

Unfolded, an element of νB over a state is the set of that state's transitions, the set of its residuals' transitions, and so on: the Aczel reading $\{(a, P') : P \xrightarrow{a} P'\}$ with Name -valued labels and an abstraction summand for input. A loop is non-well-founded descent under the colour-blind $\in_*$ of [2], §7.2 — a red process contains a black name whose red unfolding contains a red process, without end. The construction's global non-well-foundedness *is* the recursion of process behaviour, available for free where Aczel postulates it.

5 The RSpace denotation: parallel composition as keyed multiset union

The commitment reading of the previous functor invites a sharper denotation. Modulo $\equiv$ the parallel fragment $(\mid, \mathbf{0})$ is the free commutative monoid, and the free commutative monoid is the finite-multiset construction. Interpreting $\mid$ as multiset union therefore makes $\llbracket - \rrbracket$ a monoid homomorphism *on the nose*, and structural congruence of the parallel skeleton becomes literal equality of multisets — the tightest fit syntax and semantics can have. Bucketing the multiset by channel turns it into a hashtable; this is exactly the RSpace store, and the denotation below is, term for term, the compilation of rho into RSpace states.

5.1 The table functor

Definition 11 (RSpace tables). A *table* is a finite partial map from names to *polarity-homogeneous* multisets of payloads,

$$\mathsf{T}(R) = \left(\text{Name} \rightarrow \left(\mathcal{M}_{\text{fin}}(R) \uplus \mathcal{M}_{\text{fin}}([\mathcal{N}]R) \right) \right),$$

where each key lands in exactly one summand: a key holds either a multiset of *output payloads* $x \mapsto \{ \langle Q_1, \dots, Q_m \rangle \}^+$ (produces, data) or a multiset of *input continuations* $x \mapsto \{ \langle (y_1) P_1, \dots, (y_n) P_n \rangle \}^-$ (consumes, abstractions), never a mixture. $\mathcal{M}_{\text{fin}}$ is the finite-multiset (free commutative monoid) functor; $[\mathcal{N}]R$ is the name-abstraction of [2], §6.2. T acts on the knotted nominal universe $\mathcal{U}$, the keys being names (red atoms) and hence, by Section 3, quoted processes — content-addressed channels.

Definition 12 (The RSpace denotation). Define $\llbracket - \rrbracket : \text{Proc} \rightarrow \mathsf{T}(\text{Proc})$ by

$$\begin{aligned} \llbracket \mathbf{0} \rrbracket &= \emptyset, & \llbracket x ! (Q) \rrbracket &= \{ \langle x \rangle \mapsto \{ \langle Q \rangle \}^+ \}, & \llbracket \text{for}(y \leftarrow x) P \rrbracket &= \{ \langle x \rangle \mapsto \{ \langle (y) \langle P \rangle \rangle \}^- \}, \\ \llbracket P \mid Q \rrbracket &= \llbracket P \rrbracket \uplus \llbracket Q \rrbracket, & \llbracket *x \rrbracket &= * \langle x \rangle = \mathsf{s}(\kappa^{-1} \langle x \rangle), & \llbracket @P \rrbracket &= @ \llbracket P \rrbracket = \kappa(\mathsf{s} \llbracket P \rrbracket), \end{aligned}$$

where $\uplus$ is *key-wise* multiset union: $(\llbracket P \rrbracket \uplus \llbracket Q \rrbracket)(x)$ is the multiset union of the bags P and Q install at x . Each prefix installs a singleton bag; $\mid$ co-locates produces and consumes in the store. A key whose two operands carry the same polarity stays homogeneous; a key receiving both polarities is a *redex* (Definition 14).

Proposition 13 (Exact homomorphism). $\llbracket - \rrbracket$ is a homomorphism of commutative monoids: $\llbracket \mathbf{0} \rrbracket = \emptyset$ and $\llbracket P \mid Q \rrbracket = \llbracket P \rrbracket \uplus \llbracket Q \rrbracket$, with $\uplus$ associative and commutative and $\emptyset$ its unit. Hence the parallel laws of $\equiv$ — associativity, commutativity, unit — hold as literal equalities of tables, and $P \equiv Q \Rightarrow \llbracket P \rrbracket = \llbracket Q \rrbracket$. With the reflection law of Proposition 6, every generator of $\equiv$ is validated.

Proof sketch. $\uplus$ inherits the commutative-monoid laws of multiset union key-wise; the empty table is the unit. The prefix clauses are singletons, so reassociating or commuting a parallel composition reassociates or commutes the corresponding bag unions, which multiset equality absorbs. This is the gain over the corecursive $\parallel$ of the set model: multiset union has no idempotence to defeat, so $a ! (0) \mid a ! (0)$ installs $\{ \langle \langle 0 \rangle, \langle 0 \rangle \rangle \}^+$ at a , a two-element bag, distinct from the one-element bag of $a ! (0)$ — counts are kept, and the operation is union outright rather than a context-carrying expansion. $\square$

5.2 Eager reaction is the zip, and the swap fires

Definition 14 (Eager reaction). A key x carrying both a produce-bag $B^+ = \{\{Q_1, \dots, Q_m\}^+\}$ and a consume-bag $B^- = \{\{(y_1) P_1, \dots, (y_n) P_n\}^-\}$ is a redex. Eager reaction fires it: choose underlying lists of B^+ and B^- and zip them; each matched pair reacts by application,

$$(y_i) P_i \cdot Q_i = P_i\{\text{@}Q_i / y_i\} = P_i\{\kappa(\mathbf{s} Q_i) / y_i\},$$

and the resulting process is merged, by $\uplus$, back into the table; the $|m - n|$ unmatched payloads of the longer bag remain at x , now a homogeneous surplus (and x is dropped from the table when $m = n$). Eager reaction fires every redex, including those created by merged reaction products; its *normal forms* are the tables with no mixed-polarity key. The seal $\text{@} = \kappa \circ \mathbf{s}$ acts in each application: *communication is where the colour-swap of Section 3 fires*, delivering the name $\text{@}Q_i$ of the sent process, not the process.

Remark 15 (The pairing is COMM’s nondeterminism, located). Which Q_i meets which $(y_j) P_j$, and which payloads survive as surplus, depend on the chosen list orderings: this is precisely rho’s choice of which send meets which receive, now localised at the key. Eager reaction is therefore a *relation* on tables, confluent only up to the equivalence that forgets the pairing — resource bisimilarity (Section 6). The denotation $\llbracket - \rrbracket$ remains a function (plain $\uplus$, no firing); reaction is the separate dynamics. The minimal context $[\cdot] \mid x!(Q)$ of Definition 1 is exactly the minimal store-insertion — one produce at key x — that turns a pending consume into a redex; context labels are RSpace puts.

5.3 RSpace as a bialgebra, and the factoring

Definition 16 (The RSpace bialgebra). The table functor T carries a monad structure: $\mathcal{M}_{\text{fin}}$ is the free commutative-monoid monad, and the key-wise lift $\mathbf{Name} \rightarrow \mathcal{M}_{\text{fin}}(-)$ is the store, with multiplication $\uplus$. RSpace is this monad *together with* the eager reaction of Definition 14, presented as a distributive law of the reaction over the store — a bialgebra in the sense of Turi-Plotkin [16]. This makes precise the standing description of RSpace as “strictly more than a monad”: the monad is the store ($\mathcal{M}_{\text{fin}}$ -valued map, for-comprehension as the consume bag), and the surplus is the distributive law — the zip, with send-and-substitute $P_i\{\text{@}Q_i / y_i\}$ as its witness.

Proposition 17 (The factoring onto $\nu\mathsf{B}$). *Let $\text{supp}: \mathcal{M}_{\text{fin}} \Rightarrow \mathcal{P}$ be the support transformation (forget multiplicities) and let flat flatten a table into its set of context-labelled transitions (a consume at x is an offer-input transition, a produce an offer-output, a mixed key an internal redex). The composite $\pi = \text{flat} \circ \text{supp}: \mathsf{T}(R) \rightarrow \mathsf{B}(R)$ is a natural transformation, and it carries the RSpace dynamics to the context-labelled system of Section 2.3. Hence the RSpace table model sits above the behaviour model and projects onto it:*

$$\text{Syntax} \xrightarrow{\llbracket - \rrbracket} \mathsf{T}\text{-tables (RSpace)} \xrightarrow{\pi} \nu\mathsf{B},$$

and the composite is the context-behaviour denotation of Section 4. Resource bisimilarity (Definition 19) refines context bisimilarity, with π the quotient that forgets multiplicity and keys.

The reflection law is undisturbed by the refinement — it concerns names, not the store monoid — so Theorem 18 below holds verbatim, now with the homomorphism into tables.

Theorem 18 (Adequacy and the reflection law). *For all processes P, Q and names x :*

- (i) $\langle\!\langle @P \rangle\!\rangle = @\langle\!\langle P \rangle\!\rangle$, $\langle\!\langle *x \rangle\!\rangle = *x$, and $\langle\!\langle *\langle\!\langle P \rangle\!\rangle \rangle\!\rangle = \langle\!\langle P \rangle\!\rangle$: the reflection law holds in the model, as $s \circ s = \text{id}$ (Proposition 6);
- (ii) $\langle\!\langle - \rangle\!\rangle$ is a monoid homomorphism into tables (Proposition 13); in particular $P \equiv Q \Rightarrow \langle\!\langle P \rangle\!\rangle = \langle\!\langle Q \rangle\!\rangle$, the parallel-monoid laws holding as literal table equalities and the reflection law as the involutivity of s .

6 Full abstraction

The factoring of Proposition 17 gives two full-abstraction statements, one at each layer. At the *resource* layer the model counts parallel copies, so the equivalence it realises is resource bisimilarity; at the *behavioural* layer, reached by the projection π that forgets multiplicity and keys, it is the context bisimilarity of Section 2.3. We give the resource statement first, then recover the behavioural one as its image under π .

6.1 The resource layer

Definition 19 (Resource bisimilarity). A symmetric relation $\mathcal{R}$ on tables is a *resource bisimulation* if related tables carry, key by key, multisets matched *with multiplicity*: equal counts of produces and of consumes at each channel, with $\mathcal{R}$ -related payloads and (after the eager zip of Definition 14) $\mathcal{R}$ -related residuals. Resource bisimilarity $\sim_r$ is the largest such relation; on processes, $P \sim_r Q$ means $\langle\!\langle P \rangle\!\rangle \sim_r \langle\!\langle Q \rangle\!\rangle$. It counts copies: $a!(0) \mid a!(0) \not\sim_r a!(0)$.

Lemma 20 (The store functor is well behaved). *The finite-multiset functor $\mathcal{M}_{\text{fin}}$ preserves weak pullbacks, because $(\mathbb{N}, +)$ is a refinement monoid (the Riesz interpolation property) [11]; key-wise lifting, products, coproducts and the abstraction functor preserve weak pullbacks too, so the table functor T preserves them. Hence on any T -coalgebra, resource bisimilarity is the kernel of the unique morphism into the final table-coalgebra $\nu\mathsf{T}$.*

Theorem 21 (Resource full abstraction). *For all rho processes P, Q ,*

$$\langle\!\langle P \rangle\!\rangle \sim_r \langle\!\langle Q \rangle\!\rangle \iff P \sim_r Q.$$

Equivalently, $\langle\!\langle - \rangle\!\rangle$ followed by the unique morphism into $\nu\mathsf{T}$ identifies exactly the resource-bisimilar processes. Replication forces a multiplicity of ω ($!P \equiv !P \mid P$ installs ω copies), so $\nu\mathsf{T}$ is taken with multiplicities in $\mathbb{N} \cup \{\omega\}$ — and that ω is the same one the risk ledger (Section 7) charges as black infinity.

Proof sketch. $\langle\!\langle - \rangle\!\rangle$ is a homomorphism into tables (Proposition 13); equipping $\mathcal{T}/\equiv$ with the eager dynamics makes it a T -coalgebra, and the unique morphism into $\nu\mathsf{T}$ exists by Lemma 20. Its kernel is resource bisimilarity by the same lemma; that the syntactic resource bisimilarity $\sim_r$ matches the coalgebraic one is the multiset analogue of Lemma 22, the offer-input and offer-output families now compared with multiplicity. The ω -multiplicity refinement is the only point where infinitude enters, and it enters exactly once. $\square$

6.2 The behavioural layer, by projection

Applying the projection π of Proposition 17 — forget multiplicity ($\mathcal{M}_{\text{fin}} \Rightarrow \mathcal{P}$) and keys — carries the resource model onto the context-behaviour model, and resource bisimilarity onto context bisimilarity. The remaining lemmas establish the behavioural statement directly, over the commitment refinement $\mathsf{B}^b = \pi \circ \mathsf{T}$.

Lemma 22 (Context = coalgebraic bisimilarity). *Context bisimilarity (Definition 3) coincides with B^b -coalgebraic bisimilarity. For the relational lifting $\overline{B^b}$, a relation $\mathcal{R}$ on $\mathcal{T}/\equiv$ is a coalgebra bisimulation iff it is a context bisimulation.*

Proof sketch. By Remark 5 the commitment presentation packages each class-indexed context-transition family as a single abstraction or concretion, with no loss of discriminating information: two states pass the same context labels iff their commitment sets match. $\overline{B^b}$ relates behaviours when their output and silent summands match with $\mathcal{R}$ -related residuals and their input summands match in $[\mathcal{N}]\mathcal{R}$; unfolding $[\mathcal{N}]\mathcal{R}$ by its defining equivalence $((y) R [\mathcal{N}]\mathcal{R} (y) R' \text{ iff } R\{a/y\} \mathcal{R} R'\{a/y\})$ for all names a) reproduces the matching of the offer-input context family, and the concretion case the offer-output family. Hence the two notions of bisimulation coincide. $\square$

Lemma 23 (B^b is well behaved). *B^b preserves weak pullbacks. Hence on any B^b -coalgebra (C, γ) , coalgebraic bisimilarity is the kernel of the unique morphism $!_C: C \rightarrow \nu B^b$: $c \sim_{B^b} d \iff !_C(c) = !_C(d)$.*

Proof sketch. $\mathcal{P}_{\text{fin}}$, products, coproducts and constants preserve weak pullbacks, and so does the nominal abstraction functor $[\mathcal{N}](-)$ (its quotient by the freshness equivalence is by a functional, hence pullback-stable, relation); weak-pullback preservation is closed under these constructions, so B^b preserves them. For a weak-pullback-preserving functor with a final coalgebra, behavioural equivalence (the kernel of $!_C$) coincides with coalgebraic bisimilarity — the standard fact of universal coalgebra [10]. Existence of νB^b , and its agreement with the class-sized νB on equality, is Obligation 1. $\square$

Theorem 24 (Full abstraction for context bisimilarity). *For all rho processes P, Q , writing $\langle\!\langle - \rangle\!\rangle_B = \pi \circ \langle\!\langle - \rangle\!\rangle$ for the behavioural denotation (the table projected to νB^b),*

$$\langle\!\langle P \rangle\!\rangle_B = \langle\!\langle Q \rangle\!\rangle_B \iff P \sim Q.$$

Moreover, since $\sim$ is a congruence (Proposition 4), this is full abstraction with respect to a congruence: the behavioural denotation identifies P and Q exactly when no context can tell them apart. It is the image under π of the resource statement (Theorem 21): $P \sim_r Q$ implies $P \sim Q$, the converse failing exactly where multiplicities differ.

Proof. By Proposition 17, $\langle\!\langle - \rangle\!\rangle_B = \pi \circ \langle\!\langle - \rangle\!\rangle$ is the unique B^b -coalgebra morphism $(\mathcal{T}/\equiv, \partial) \rightarrow \nu B^b$, so $\langle\!\langle P \rangle\!\rangle_B = \langle\!\langle Q \rangle\!\rangle_B$ iff P, Q have the same image. By Lemma 23 that image-equality is coalgebraic bisimilarity, and by Lemma 22 that is context bisimilarity $\sim$. Chaining, $\langle\!\langle P \rangle\!\rangle_B = \langle\!\langle Q \rangle\!\rangle_B \iff P \sim Q$. Congruence is Proposition 4; the projection π collapsing multiplicity gives $P \sim_r Q \Rightarrow P \sim Q$. $\square$

Corollary 25 (Adequacy through OSLF). *Let $\mathcal{L}$ be the Hennessy–Milner logic the OSLF construction generates from the context-labelled transition system, with modalities $\langle F \rangle \varphi$ indexed by the labels ∂T_K (Remark 2). By OSLF adequacy [7], $P \sim Q$ iff P and Q satisfy the same formulae of $\mathcal{L}$. Combining with Theorem 24,*

$$\langle\!\langle P \rangle\!\rangle_B = \langle\!\langle Q \rangle\!\rangle_B \iff P \sim Q \iff \forall \varphi \in \mathcal{L}. (P \models \varphi \Leftrightarrow Q \models \varphi).$$

Behavioural equality in the model, context bisimilarity, and modal indistinguishability are one relation seen three ways — the Firing, Placing, and Witnessing readings of the single label set ∂T_K — with resource bisimilarity $\sim_r$ the finer relation living one layer up.

Corollary 26 (Internal full abstraction; Aczel as a fragment). *On $\nu\mathbf{B}^b = \mathbf{Proc}$ itself, two elements are context-bisimilar iff equal: behavioural equivalence is the equality of the model. Aczel’s strong extensionality for CCS is recovered as the special case where the labels ∂T_K degenerate to an opaque set of action names over which no further context structure is observed — the equivariant fragment of Section 7.*

What the context labels bought. An action-labelled presentation would here have to negotiate congruence as an afterthought: strong bisimilarity in name-passing calculi may fail to be preserved by input prefix unless one closes under name instantiation, replacing $\sim$ by the largest bisimulation-congruence $\sim^c$. Labelling by contexts removes the negotiation. The bisimulation game of Definition 3 already ranges over every environment, so Proposition 4 delivers congruence directly (granted the relative pushouts of Obligation 5), and Theorem 24 is full abstraction for a congruence with no closure step. Rho’s structure makes the relative-pushout hypothesis especially plausible: there is no ν , so no scope-extrusion interaction with the cut, and a name is structured data, so name-for-name substitution is set-theoretic substitution under $\kappa \circ s$ with no fresh-name side conditions.

7 The risk ledger

The accounting of [2] transfers, and locates where rho’s recursion is paid for.

Finite processes: the hereditarily-finite core, no infinite risk. A process with a finite synchronisation tree is a hereditarily-finite red set with finitely many black-set atoms (its free names), living in the finitary core of [2], §5.5, where the final coalgebra of the finite-powerset functor is literal and exists at set size. Foundation and finite support are theorems, not axioms; the commitment is the black empty set and one successor, no more.

Recursion: one ω , black infinity, no primitive names. Recursion enters rho through reflection (Section 2): a finite term can loop by quoting and dereferencing its own continuation. Such a process is a non-well-founded inhabitant of $\nu\mathbf{B}$, available only in the infinitary reading, which by [2], §5.6 and §8.1 costs exactly one ω — black infinity — and lives, for the full powerset, in the class-sized final-coalgebra framework. As with the completed datatype Λ of [2], §6.2, each finite-state recursive process is set-sized while the universe of all processes is the one- ω object. No primitive name is ever reintroduced; names remain black sets built from $\emptyset$. The RSpace layer sharpens the accounting: replication $!P \equiv !P \mid P$ installs an ω -fold bag at a key, so the same single ω appears as a multiplicity in $\mathbb{N} \cup \{\omega\}$ (Theorem 21). Infinitude shows up once, in the count, and it is black infinity.

Aczel’s CCS model as the equivariant shadow. Stay within one colour, never apply s to a name’s interior, treat names as opaque swap-only atoms (the equivariant fragment, [2], §6.5). The input summand’s abstraction degenerates, $\mathbf{B}$ reduces to $\mathbf{B}_0(R) = \mathcal{P}_{\text{fin}}(\text{Act} \times R)$, and $\nu\mathbf{B}_0$ is Aczel’s CCS model (Corollary 26). Aczel’s construction is the colour-respecting fragment; the rho surplus — structured, computed, content-addressed names, and the licence to cross the colour barrier with $*$ — is the reflective remainder, graded by the four sorts of Section 3.

8 Toward a rigorous construction

In the manner of [2], §8, the obligations a fully rigorous development would discharge. For the finitary core each reduces to a statement about hereditarily-finite sets, and several are corollaries of the model built.

Obligation 1 (Existence of the final coalgebra, and $\nu\mathbf{B}^b \sim \nu\mathbf{B}$). Exhibit the final coalgebras via the two-sorted endofunctor $H(R, B) = (B + \mathcal{P}_s R, R + \mathcal{P}_s B)$ of [2], §8.1, taken at the *final* coalgebra (greatest fixed point). For the commitment refinement $\mathbf{B}^b$, finite branching makes it accessible and $\nu\mathbf{B}^b$ exists in the nominal universe, at set size on the finitary core (the limit of the terminal ω^{op} -chain). For the full context functor $\mathbf{B} = \mathcal{P}(\partial T_K \times -)$, the final coalgebra is the class-sized, colour-blind non-well-founded universe of [2], §7. Show the two carry the same equality — that the commitment packaging of Remark 5 is a behavioural equivalence — so bisimilarity computed in $\nu\mathbf{B}^b$ is context bisimilarity in $\nu\mathbf{B}$.

Obligation 2 (The RSpace bialgebra: eager confluence and store coherence). Make the RSpace layer of Section 5 rigorous. (i) Prove eager reaction (Definition 14) confluent up to resource bisimilarity: the pairing nondeterminism of Remark 15 affects which payloads meet, but all maximal reaction sequences reach $\sim_r$ -equal normal forms. (ii) Establish the distributive law of Definition 16 and its bialgebra laws, so “RSpace = store monad + zip” is a theorem in the Turi–Plotkin format [16], with $(-)$ the induced bialgebra morphism. (iii) Confirm $\mathbf{T}$ preserves weak pullbacks in the *nominal* and non-well-founded setting (Lemma 20 gives it over $\mathbb{N}$ by refinement [11]; the $\mathbb{N} \cup \{\omega\}$ completion forced by replication must be checked to retain the refinement property).

Obligation 3 (Non-well-founded keyed multisets in the knotted universe). Encode tables inside the knotted universe so that recursion is accommodated: a table is a red set of triples $\langle \text{key}, \text{count}, \text{payload} \rangle$ with black-set keys (red atoms) and red-numeral counts, so a non-well-founded bag is a non-well-founded set of triples — the non-well-foundedness living in the payload, the multiplicity remaining a numeral. Verify that the swap $\mathbf{s}$ and the knots $\kappa, \bar{\kappa}$ respect the count arithmetic, and that multiplicity- ω is exactly the black infinity of Section 7.

Obligation 4 (The stratified signature and its coherence). Fix the four-sorted signature of Section 3 — the sorts $\mathbf{S}, \mathbf{A}, \mathbf{S}, \mathbf{A}$, the knots $\kappa, \bar{\kappa}$, the swap $\mathbf{s}$ — as an extension of the $\text{Set}[X]$ signature of [2], and verify the equational theory: $\mathbf{s}$ an involution and a homomorphism for all set operations, $\mathbf{s}\kappa = \bar{\kappa}\mathbf{s}$, and $\kappa, \bar{\kappa}$ isomorphisms. Then $@, *$ are derived and Proposition 6 is a calculation. The collapse of [2] — $\kappa, \bar{\kappa}$ literal identities — is recovered as a quotient of this theory, and one checks the reflection law survives the quotient.

Obligation 5 (Relative pushouts for rho’s contexts). Discharge the hypothesis of Proposition 4: that the category of rho contexts, with plugging as composition, possesses relative pushouts, so the idem-pushout labels of Definition 1 are well defined and the Leifer–Milner congruence theorem applies. This is the GSLT minimal-context construction [7]; the cut lying always at a parallel composition (no reaction under a prefix), and names being quoted structure rather than ν -bound, are the features that should make the relative pushouts exist and be computable. With this discharged, $\sim$ is a congruence outright, and the action-labelled closure-under-substitution question does not arise.

Obligation 6 (Context correspondence and OSLF adequacy). Promote Lemma 22 to a theorem: define the relational lifting $\overline{\mathbf{B}^b}$ precisely, including the abstraction summand, and prove context bisimilarity equals $\overline{\mathbf{B}^b}$ -bisimilarity. Relate $\sim$ to reduction-barbed congruence so that Theorem 24 is

full abstraction for the intended equivalence; rho’s lack of ν removes the closure under restriction this normally requires. Finally, instantiate the OSLF construction on the context-labelled system and prove the adequacy used in Corollary 25, so that the modalities $\langle F \rangle \varphi$ indexed by ∂T_K characterise $\sim$.

Obligation 7 (Substitution and the swap-firing reaction). Prove $\langle P\{\textcircled{M}/y\} \rangle = \langle P \rangle \{\textcircled{M}/y\}$, i.e. that $\langle - \rangle$ is a map of nominal algebras for substitution and that the application of Definition 14 computes it, with the seal $\kappa \circ \mathbf{s}$ applied to the message. This is the lemma underwriting the eager zip and clause (ii) of Theorem 18 for input, carried by the abstraction functor’s evaluation $[\mathcal{N}]R \times \mathbf{Name} \rightarrow R$.

Obligation 8 (Barbed-congruence calibration). With congruence secured by Obligation 5, calibrate which behavioural equivalence the model realises: confirm that context bisimilarity $\sim$ coincides with strong barbed congruence for rho, so that Theorem 24 is full abstraction for the equivalence practitioners use. Rho’s structured, ν -free names are again the reason to expect the calibration to be tight where in π it is delicate.

Obligation 9 (Placing the construction). Locate the model among Aczel’s CCS model [1] (the equivariant fragment), the presheaf and nominal coalgebraic models of name-passing (Fiore–Turi [14], Stark [15]), of which this is the reflective, ν -free instance with $\mathbf{Name} \cong \mathbf{Proc}$, and the bialgebraic semantics of Turi–Plotkin [16], into whose abstract-GSOS format the rho rules fit. The defining feature absent from all three — $\mathbf{Name} \cong \mathbf{Proc}$ via $\mathbf{s}$ — is the second knot.

9 Conclusion

Rho’s defining move — names are quoted processes, $*\textcircled{P} \equiv P$ — is the colour-swap of the knotted universe, read through a stratification into four sorts: $\textcircled{} = \kappa \circ \mathbf{s}$, $* = \mathbf{s} \circ \kappa^{-1}$, and the reflection law is the involutivity $\mathbf{s} \circ \mathbf{s} = \text{id}$, true by construction. Parallel composition is interpreted as keyed multiset union, which makes the denotation a monoid homomorphism on the nose — the parallel laws of structural congruence become literal equalities of tables — and the table *is* the RSpace store: a hashtable from channels to polarity-homogeneous bags of produces and consumes. Communication is eager reaction, the zip of dual bags at a key, with the colour-swap firing to seal each sent process into the received name; RSpace is the store monad together with this zip, a bialgebra, which is the precise content of “strictly more than a monad”. Following the GSLT discipline we labelled transitions by contexts — the minimal store-insertions that enable a reaction, which are at once the program/environment cuts, the OSLF modality indices, and the locations in term structure — and expressed bisimulation over them. The model factors: the resource layer counts copies and realises resource bisimilarity; the projection that forgets multiplicity and keys lands on the context-behaviour layer, where we proved full abstraction, $\langle P \rangle_{\mathbf{B}} = \langle Q \rangle_{\mathbf{B}} \iff P \sim Q$, for a congruence with no closure step. The risk ledger is the source paper’s: finite processes cost nothing beyond the empty set, recursion costs one ω — now visible as an ω -fold multiplicity — and not one primitive name is ever spent. Aczel’s model sits inside this one as its equivariant shadow; the surplus is reflection, and reflection is the swap.

Future work. Three threads we have deliberately set aside. The reaction strategy here is *eager*, with tables reduced to polarity-homogeneous normal forms; the *lazy* reading, treating a mixed-polarity key as a standing redex and reacting on schedule, is closer to the coalgebraic machinery and deserves its own treatment. The polarity-homogeneity invariant of reduced tables — a key

holds produces or consumes, never both — carries a proof-theoretic reading we leave for a separate study. And the two involutions now in play, the colour-swap on names and the polarity duality on channels, may or may not coincide; we have not pursued the question here.

Acknowledgements

This note develops Section 9.3 of *A Knotted Universe*, with the stratified reading of the colour barrier, the Milner-style commitment semantics over context-labelled transitions, and the full-abstraction theorem as its additions.

References

- [1] Peter Aczel. *Non-Well-Founded Sets*, volume 14 of CSLI Lecture Notes. CSLI Publications, Stanford, CA, 1988.
- [2] L. G. Meredith. A Knotted Universe: a new notion of reflective set theories. Manuscript, 2026.
- [3] L. Gregory Meredith and Matthias Radestock. A reflective higher-order calculus. *Electr. Notes Theor. Comput. Sci.*, 141(5):49–67, 2005.
- [4] Robin Milner. The Polyadic π -Calculus: A Tutorial. In F.L. Bauer, W. Brauer, and H. Schwichtenberg, editors, *Logic and Algebra of Specification*, volume 94 of NATO ASI Series F, pages 203–246. Springer, 1993. (Also Technical Report ECS-LFCS-91-180, University of Edinburgh, 1991.)
- [5] James J. Leifer and Robin Milner. Deriving bisimulation congruences for reactive systems. In *CONCUR 2000*, volume 1877 of Lecture Notes in Computer Science, pages 243–258. Springer, 2000.
- [6] Peter Sewell. From rewrite rules to bisimulation congruences. *Theoret. Comput. Sci.*, 274(1–2):183–230, 2002.
- [7] L. G. Meredith, Christian Wells, et al. Minimal-context transition systems and the observer-specified logic functor for graph-structured lambda theories. Manuscript, forthcoming.
- [8] Murdoch Gabbay and Andrew M. Pitts. A new approach to abstract syntax with variable binding. *Formal Asp. Comput.*, 13(3–5):341–363, 2002.
- [9] Andrew M. Pitts. *Nominal Sets: Names and Symmetry in Computer Science*, volume 57 of Cambridge Tracts in Theoretical Computer Science. Cambridge University Press, 2013.
- [10] J. J. M. M. Rutten. Universal coalgebra: a theory of systems. *Theoret. Comput. Sci.*, 249(1):3–80, 2000.
- [11] H. Peter Gumm and Tobias Schröder. Monoid-labeled transition systems. *Electron. Notes Theor. Comput. Sci.*, 44(1):185–204, 2001.
- [12] Robin Milner. *Communication and Concurrency*. Prentice Hall, 1989.
- [13] Davide Sangiorgi and David Walker. *The π -Calculus: A Theory of Mobile Processes*. Cambridge University Press, 2001.

- [14] Marcelo Fiore and Daniele Turi. Semantics of name and value passing. In *16th Annual IEEE Symposium on Logic in Computer Science (LICS 2001)*, pages 93–104. IEEE, 2001.
- [15] Ian Stark. A fully abstract domain model for the π -calculus. In *11th Annual IEEE Symposium on Logic in Computer Science (LICS 1996)*, pages 36–42. IEEE, 1996.
- [16] Daniele Turi and Gordon Plotkin. Towards a mathematical operational semantics. In *12th Annual IEEE Symposium on Logic in Computer Science (LICS 1997)*, pages 280–291. IEEE, 1997.
- [17] André Joyal and Ieke Moerdijk. *Algebraic Set Theory*, volume 220 of London Mathematical Society Lecture Note Series. Cambridge University Press, 1995.